Quiz 6

M 375T Mathematical Foundations of Machine Learning — Fall 2026

15 minutes · closed book · no devices

Name: _____ UT EID: _____

Vertices $0, 1, 2$; laws move by $\mu_{t+1} = P^{\mathsf{T}}\mu_t$.

$$P = \begin{pmatrix} 0 & \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & 0 & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} & 0 \end{pmatrix}, \qquad R = \begin{pmatrix} 0 & \frac{2}{3} & \frac{1}{3} \\ \frac{1}{3} & 0 & \frac{2}{3} \\ \frac{2}{3} & \frac{1}{3} & 0 \end{pmatrix}, \qquad \pi = \left(\tfrac{1}{3}, \tfrac{1}{3}, \tfrac{1}{3}\right)^{\mathsf{T}}.$$

(a) Write the equation "π is invariant" and the equation "detailed balance holds for π". [3 pts]

(b) Prove: detailed balance implies invariance. Does the converse hold? [3 pts]

(c) Which of the two hold for P? For R? Show the computation. [4 pts]

Answer:

Total: 10 points. Lowest two quiz scores are dropped at the end of the semester.